Status	Finished
Started	Saturday, 24 October 2026, 8:31 PM
Completed	Saturday, 24 October 2026, 8:42 PM
Duration	11 mins 42 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 89 and 99

b. 90 and 94

c. 90 and 99 ✓

d. 89 and 94

Question **2**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

a. Mining

b. Discovering

c. Finding

d. Searching ✓

Question **3**

Correct

Mark 1.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

a. The search stops

b. The middle element is chosen randomly

c. The lower middle element is chosen as the middle element

d. The higher middle element is chosen as the middle element ✓

Question **4**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. There should be direct access to the middle element in any sublist

b. Number values should only be present

Number values should only be present

c. The list must be sorted

d. There must be a mechanism to delete and/or insert elements in the list ✓

Question **5**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The list is sorted

b. The search ends successfully ✓

c. The search continues in the left sublist

d. The search continues in the right sublist

Question **6**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. The search is considered unsuccessful ✓

b. The last element is returned

c. The first element is returned

d. An error is raised

Question **7**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list is unsorted
b. When the list is very small
c. When the list is sorted ✓
d. When the list contains duplicate elements

Question **8**

Correct

Mark 1.00 out of 1.00

What is the first step in binary search?

a. Compare the target element with the first element in the list
b. Divide the list into two equal parts
c. Sort the list
d. Compare the target element with the middle element in the list ✓

Question **9**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

a. In the left sublist
b. In the middle of the list
c. In the right sublist ✓
d. At the beginning of the list

Question **10**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

a. Checking each element sequentially ✓
b. Dividing the list in half repeatedly
c. Sorting the list before searching
d. Skipping every second element

Question **11**

Correct

Mark 1.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

- a. Binary search
- b. Interpolation search
- c. Hash search
- d. Linear search ✓

Question **12**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

- a. It requires the list to be sorted ✓
- b. It does not work with negative numbers
- c. It is slower than linear search for small lists
- d. It can only be applied to large lists

Question **13**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

- a. By comparing each element in the list sequentially ✓
- b. By sorting the list first
- c. By dividing the list into halves
- d. By using a hash function

Question **14**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

- a. Merge search
- b. Linear search ✓
- c. Bubble search
- d. Quick search

Question **15**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

- a. n ✓
- b. n/2
- c. n-1
- d. log n